

Supportive information

**Challenges for critical raw material recovery from WEEE
- the case study of gallium**

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S1 Data collection

The mass flow analysis for gallium in the systems production, processing industry and waste management is based on many scattered data, which has to be harmonized. Many data sets are very difficult to access or do not even exist so that assumptions or data conversions were made based on different sources. Depending on the reliability and accuracy of original data to be transferred into the model, uncertainty must be determined and used for further calculations. Table 1 shows all required data for the development of the gallium MFA model for all investigated systems.

Table 1: Required data for the gallium MFA for each investigated system

System	Process	Required data
Production	Primary production	- Production capacity - Amount of the obtained primarily raw gallium - Losses during production
	Refining	- Input amount of raw gallium in process (from primary and secondary production) - Amount of refined gallium - Losses during refining processes
	Processing industry	- Input amount of refined gallium - Amount of gallium in finished chips for use in electrical equipment - Losses during processing - Amount of recyclable materials from production residues
	Recycling	- Input amount of gallium to be recycled - Losses during recycling - Amount of recycled gallium
	Sink	- Input streams consisting of losses of all processes in the system production
Processing industry	Wafer production	- Input amount of refined gallium - Losses due to production waste - Amount of recyclable materials from production residues - Amount of gallium finally applied in wafer
	Chip production	- Input amount of gallium in wafer - Losses during production processes - Amount of gallium in finished chips for use in electrical equipment
Waste management	Generation of WEEE	- Mass fractions of gallium in WEEE devices - Amount of gallium available for recycling through formal collection of WEEE - Amount of gallium in not collected WEEE
	Collection	- Mass of collected gallium containing WEEE devices - Amount of gallium, which enters pre-processing of WEEE
	Pre-processing of WEEE	- Input amount of gallium - Losses during pre-processing - Gallium amounts in processed and sorted metal concentrates for further recycling steps
	Further processing / end-refining	- Input amount of gallium in metal concentrates - Gallium losses during further recycling steps - Amount of gallium recovered during further processing steps / end-refining
	Sink	- Input streams from losses of all processes in the system "waste management"

S2 Data base

S2.1 Import and export of gallium

Table 2: Import quantities of gallium in Germany in 2012

Country	Import	Share of total import
	[kg]	[%]
Belgium	700	2.1
China	3,000	8.98
France	100	0.3
Great Britain	14,600	43.71
Netherlands	100	0.3
Slovakia	11,900	35.63
Ukraine	100	0.30
USA	2,900	8.68
Total	33,400	100

Based on (Liedtke, 2015)

Table 3: Export quantities of gallium in Germany in 2012

Country	Export	Share of total export
	[kg]	[%]
Australia	100	0.25
Great Britain	9,400	23.38
Italy	100	0.25
Japan	100	0.25
Korea	100	0.25
Luxembourg	100	0.25
Austria	100	0.25
Poland	1,300	3.23
Sweden	100	0.25
Switzerland	4,500	11.19
Singapore	100	0.25
Slovakia	100	0.25
Slovenia	100	0.25
USA	24,000	59.70
Total	40,200	100

Based on (Liedtke, 2015)

S2.2 Data base for system “production”

S2.2.1 Primary production of gallium

Germany represents one of the major producers of primary, secondary and generally refined gallium (USGS, 2016). INGAL Stade, which is owned by Molycorp (50 %) and 5N plus (50 %), is main producer of primary gallium in Germany (Roskill Information Services, 2014). The production capacity is 35 t/a (Liedtke, 2015). In 2012, 32 t were produced (Roskill Information Services, 2014). Only approximately one third of the actual potential of gallium in bauxite is extractable (Liedtke, 2015). Primary production processes from lye originating from the Bayer process as input material transfer 70-80 % of this extractable potential to refining processes. The remaining 20-30 % are disposed to the red mud (Zhao et al., 2012).

S2.2.2 Refining of gallium

The purity is highly important for the application of gallium in electronic goods. Raw gallium usually has a purity between 99.0 % and 99.9 %. For the further use as semi conductors in semi-finished products, purities of 99.9999 % (6N) up to 99.999999 % (8N) are needed (Ullmann’s Encyclopedia, 2012). For this, refining processes separate present impurities. The recovery of gallium chloride from acid solutions and fractional distillation of liquid gallium represents one option. Higher purities are obtained by fractional crystallization, zone melting, or single crystal growth (Løvik et al., 2015; Moskalyk, 2003; Ullmann’s Encyclopedia, 2012). Zone melting represents one of the most effective procedures (Ghosh et al., 2009). Depending on the process, various losses occur. Due to impurities, edges are cut and discarded after the zone melting process. In these residues, small amounts of gallium may be present. In the single crystal growth process, gallium can be transferred to the remaining melt. (Wittmer et al., 2011)

Germany has insufficient refining capacities. Almost the entire produced primary gallium in Stade as well as the secondary material from the processing industry in Germany are shipped abroad to the United States and the United Kingdom (Liedtke, 2015).

Only PPM Pure Metals in Langelsheim is refining gallium. In 2013, the capacity of refined gallium was 10 t/a. This value is taken for further calculations in the MFA model. An according uncertainty was determined (cf. original study). (Roskill Information Services, 2014)

Refining is related to losses, depending on the origin of the material (primary, secondary source), used refining processes, and further unspecified losses. For this, no concrete data is available. Therefore, an assumption of 1 % was adopted for this MFA, originating from (Dehnavi, 2013).

S2.2.3 Processing of refined gallium

In Germany, refined gallium is used mainly for further processing in microelectronics and to a certain extend as wafer in optoelectronics (Liedtke, 2015). Companies, that handle gallium are listed in Table 4.

Table 4: Companies processing gallium in Germany

Company	Products
Freiberger Compound Materials	Mainly GaAs wafer
Osram Opto Semiconductors	GaAs, InGaN, InGaP, Wafer for LEDs
AZZURRO Semiconductors	GaN on silicon epitaxialwafer for LEDs
BOSCH	CIGS
Centrotherm Photovoltaics	CIGS
Solarion	CIGS
Solibro (Hanergy)	CIGS
Würth Solar GmbH & Co KG	CIGS

Based on (Roskill Information Services, 2014)

One of the world's largest producers of compound semiconductor substrates for micro- and optoelectronics is the Freiberger Compound Materials (FCM) in Freiberg/Germany. In this study, the authors assume, that most of the current gallium flows enter here (Liedtke, 2015). FCM contributes a share of 37 % of the total GaAs substrates supply for the world market (Roskill Information Services, 2014).

S2.2.4 Wafer production

The refined gallium is then entering the processing industry, in which it is processed to GaAs for example by means of arsenic high-pressure synthesis followed by crystal growth. Sawing and etching processes combined with subsequent polishing and cleaning steps lead to the finished wafer. Through this, process residues arise, which are differentiated in two kind of material streams. From some of these residues, gallium can be recovered. Therefore, they represent recycling streams in this study. Gallium in other streams are irretrievable lost. 60 % of the gallium input is transferred to the process residues in the wafer production. 18 % gallium, originating from scraps from crystal growing, sawing and etching processes, and 30 % in form of mud are re-directed to recycling processes. This sums up to 48 %. The remaining 12 % are lost through waste water, respectively remains in the arsenic containing precipitation sludge from the subsequent waste water treatment. In the recycling processes, about 3 % gallium is lost through process residues. Therefore, 45 % are redirected to the wafer production process (Stelter and Zeidler, 2013). FCM covers about 40-50 % of its input with this secondary gallium.

S2.2.5 Chip production

Significant losses of gallium take place in the last step of the production chain. Process residues from substrate processing and chip production cannot be recycled to recover gallium. These residues are generated while reducing the substrate thickness after the epitaxial layers have been produced. Partially, a complete removal is conducted for the production of specific LED. Through this, only about 8 % of the total used gallium is applied in final EEE products (Stelter and Zeidler, 2013).

Wafer production takes place mainly in the production of microelectronics and to a small extend in the production of optoelectronics. Wafer, produced by FCM in Germany, are usually exported. Further processing is carried out in Asian countries. The final components are then assembled in the USA as there is no market for this in Germany (Liedtke, 2015).

OSRAM Opto Semiconductors GmbH produces semiconductors for LED mainly for own needs. But, the manufacturing of the end product takes place in Malaysia. (OSRAM Opto Semiconductors, 2015)

AZURRO semiconductors was a relatively small company with 42 employees in 2012 with headquarters in Dresden. In 2014, AZURRO stopped the production and registered their insolvency. They were specialized in the field of wafer production for LED and power electronics. (ALLOS Semiconductors, 2015)

S2.3 Data base for system “waste management”

S2.3.1 WEEE collection

The system "waste management" starts with the obsolescence of EEE. Yet, not all generated WEEE is collected with formal collection systems. The calculation of these losses bases on extrapolations and reasonable assumptions. Due to a general lack of data, figures from 2007 are used and might differ to the data in 2012 (Chancerel, 2010). Comparisons for single devices for different years revealed only low differences (Manhart et al., 2012). Differences to the actual collection data gives indication for losses (stiftung elektro-altgeräte register, 2015).

In Germany, WEEE is collected and grouped in collection groups (CG). The CG and the related collected masses in Germany in 2012 are displayed in Table 5.

Table 5: Collection of CG 3 and 5 in Germany in 2012

Collection group	Device categories	Put-on-market	Collection			
			Private households	Other sources	Total category	Total Collection group
		[t]	[t]	[t]	[t]	[t]
CG 3	IT and telecommunication	248,878	137,045	23,080	160,125	331,479
CG 3	Entertainment electronics	180,767	160,299	11,055	171,354	
CG 5	Lighting products without gas discharge lamps	67,711	445	136	581	10,260
CG 5	Gas discharge lamps		8,455	1,224	9,679	

Based on (Umweltbundesamt, 2015)

The CG consist of various device types. Table 6 shows the clusters of further considered devices per CG.

Table 6: Clustering of investigated devices

Equipment group	Considered devices
Mobile phones	Mobile phones
Desktop PC	Desktop PC
CRT monitor	-
Large high-grade equipment	DVD player, video games / consoles, TFT monitors, notebooks PC, Printer, VCR
Small high-grade equipment	Digital cameras, camcorder
Low-grade equipment	Radio / recorder

Based on (Chancerel, 2010)

The share of the considered devices was deducted from various sources. Table 7 depicts the shares of VCR, DVD-player and radios in CG 3. Table 8 shows the shares of printers, digital cameras, camcorder and video games / consoles in CG 3 and CG 5.

Table 7: Share of VCR, DVD-player and radio / recorder in CG 3

Device	Collected units	CG 3 total	Share of CG 3
	[-]	[t]	[%]
VCR	7,300	1.936	1
DVD-Player	760		0.1
Radio/Recorder	9,700		1.7

Based on (Technische Universität Berlin, 2014)

Table 8: Share of electrical appliances in SG 3 and SG 5

Device	Share of CG 3	Share of SG 5
	[%]	[%]
Printer	6,1	
Digital camera	0,1	-
Camcorder		
Video games / consoles	0,51	

Based on (stiftung elektro-altgeräte register, 2015; Technische Universität Berlin, 2014)

For further calculations, the average unit weights are needed. Table 9 shows the average masses of the investigated devices.

Table 9: Weight mapping per piece depending on the device type

Device	UNU key	Average weight
		[kg/unit]
Desktop PC	0302	8.8
Notebook PC	0303	3
VCR	0404	2.7
DVD player	0404	2.7
Radio / recorder	0403	3.4
Printer	0304	10
Mobile phone	0306	0.1
Digital camera	0406	0.3
Camcorder	0406	0.3
Video games / consoles	0702	1.9

Based on (Baldé et al., 2015)

Gallium in PCB containing devices

Oguchi et al., 2011 examined the share of PCB in various WEEE devices. The mean values are used for further calculations. Table 10 shows the share of PCB for each device and the according gallium mass fraction in the PCB.

Table 10: Share of printed circuit boards and their levels of gallium in WEEE

Device	Share of PCB	Mass fraction of gallium in PCB
	[%]	[mg/kg]
Desktop PC (n=5)	11.7	11
Notebook PC (n=10)	15.6	10
VCR (n=18)	18.7	9
DVD player (n=4)	14.1	9
Radio / recorder (n=18)	10.4	12
Mobile phone (n=16)	29.5	140
Digital camera (n=2)	20.3	15
Camcorder (n=2)	17.7	52
Video games (n=2)	20.6	16

The equation used for the calculation of gallium losses for PCB bearing devices is depicted in Equation 1.

Equation 1: Calculation of gallium losses for PCB bearing devices

$$Ga_{coll.}[kg] = \sum mCG[kg] * S_{\frac{Device}{CG}}[\%] * S_{\frac{PCB}{Device}}[\%] * S_{\frac{Ga}{PCB}}[\frac{mg}{kg}]$$

With:

$Ga_{coll.}$ = Gallium collected with Ga containing screen devices in kg; mCG = mass of WEEE per collection group in kg; $S_{Device/CG}$ = share of PCB containing devices in collection group in %; $S_{PCB/Device}$ = share of PCB in device in %; $S_{Ga/PCB}$ = mass fraction of gallium in PCB in mg/kg

Gallium in LED containing devices

The calculation of gallium in LED bearing devices considers TFT TV, PC monitors and notebooks. The collected masses in 2013 are depicted in Table 11.

Table 11: Collected amounts of screen devices in CG 3 in Germany in 2013

Device	Collected mass
	[t]
TFT TV	5,000
PC monitors	7,500
Notebooks	3,000

Based on (Rotter et al., 2015)

An estimation of the number of applied LED per device builds the basis for the calculation of the total gallium masses per device. Only white LED are considered, as gallium is mostly used for this type (Buchert et al., 2012). In 2010, in about 30 % of LCD TV and monitors as well as in 90 % of notebooks, LED were applied for background lighting (Buchert et al., 2012). Table 12 shows the estimated LED amounts of LED per device.

Table 12: Estimated LED amounts per device

Device	Amount of white LED per device	Amount of gallium in semiconductor chip of one LED
	[-]	[μg]
Notebooks	50	32.5
PC monitors	100	
TFT TV	150	

Based on (Buchert et al., 2012)

The equation used for the calculation of gallium losses for LED bearing devices is depicted in Equation 2.

Equation 2: Calculation of gallium losses for LED bearing devices

$$Ga_{coll.}[kg] = \sum \frac{mSD_{per\ type}[kg] * S_{Devices\ with\ LED}[\%]}{m_{Device}[kg]} * \#LED * mGa_{LED}[kg]$$

With:

$Ga_{coll.}$ = Gallium collected with Ga containing screen devices in kg; $mSD_{per\ type}$ = Mass of all collected screen devices per type in kg; $S_{Devices\ with\ LED}$ = Share of devices using LED as background lighting in %; m_{Device} = Average weight of screen device in kg; $\#LED$ = Number of white LED per device; mGa_{LED} = Gallium load per LED in [kg]

Gallium losses in WEEE collection

With this data basis, gallium masses related to PCB and LED in WEEE is calculated. Table 13 shows the resulting losses of gallium due to not collected WEEE related to PCB.

Table 13: Losses of gallium due to not collected WEEE containing PCB

Device	Not collected masses	Not collected units	Not collected gallium masses	Share
	[t]	[-]	[kg]	[%]
Desktop PC	22,624	2,570,954	29	8.7
Notebook PC	2,891	963,757	4,5	1.4
VCR	3,267	1,209,953	55	16.5
DVD player	339	125,600	0,5	0.2
Radio / recorder	-1,517	-446,239	-1.9	-0.6
Printer	20,091	2,009,054	4,8	1.4
Mobile phone	6,808	68,083,682	280,8	84.3
Digital camera	775	2,582,991	1.2	1.1
Camcorder			3.6	
Video games / console	1,680	884,053	5.5	1.7
TV	4,819	892,867,173	-	-
Total	61,777	78,876,173	333	100

Table 14 shows the not collected gallium masses for LED containing WEEE.

Table 14: Losses of gallium due to not collected WEEE containing LED in background lighting

Device	Not collected masses	Not collected units	Amount of not collected LED	Not collected gallium masses	Share
	[t]	[-]	[-]	[kg]	[%]
TFT TV	4,819	892,367	267,710	1.3	25
PC monitor	22,624	2,570,954	771,286	2.5	48
Notebook	2,891	963,757	867,381	1.4	27
Total	30,334	1,856,199	1,906,377	5.2	100

The overall calculated share of collected and not collected WEEE in Germany in 2007 is shown in Table 15.

Table 15: Calculated share of collected and not collected WEEE in Germany in 2007

Equipment group	Generated WEEE	Collected	Not collected
	[t]	[t]	[%]
Mobile phones	2.273	239	89
Desktop PC	39.957	9.948	75
CRT monitor	200.000	134.126	33
Large high-grade equipment	87.071	44.339	49
Small high-grade equipment	2.550	764	70
Low-grade equipment	93.513	127.938	-37
Total	425.364	317.354	25

S2.3.2 WEEE pre-processing

WEEE processing considers dismantling and mechanical treatment. Chancerel et al., 2010 examined the routes and fates of gold and palladium in WEEE processing. As gallium is applied mostly in the same devices, the transfer coefficients were used for this study. Table 16 depicts the losses of precious metals in WEEE pre-processing

Table 16: Losses of precious metals in WEEE pre-processing

Equipment group	Losses
	[%]
Mobile phones	39
Desktop PC	40
CRT monitor	40
Large high-grade equipment	60
Small high-grade equipment	60
Low-grade equipment	71

Based on (Chancerel et al., 2010)

Gallium losses in pre-processing of PCB containing devices

Table 17: Gallium losses in pre-processing of PCB containing devices in Germany in 2012

Device	Lost gallium
	[kg]
Desktop PC	3.9
Notebook PC	2.8
VCR	3.4
DVD player	0.3
Radio / recorder	5
Printer	3
Mobile phone	12.9
Digital camera	0.3
Camcorder	0.9
Video games / console	3.5
TV	-
Total	35.8

Gallium losses in pre-processing of LED containing devices

Table 18: Gallium losses due to pre-processing of LED containing devices in Germany in 2012

Device	Lost gallium
	[kg]
TFT TV	1.4
PC monitors	0.8
Notebooks	1.5
Total	3.7

S2.3.3 Further WEEE processing

Table 19: Gallium losses in further recycling steps of PCB in Germany in 2012

Device	Lost of gallium
	[kg]
Desktop PC	5.8
Notebook PC	1.9
VCR	2.3
DVD player	0.2
Radio / recorder	2.0
Printer	2.0
Mobile phone	20.1
Digital camera	0.2
Camcorder	0.6
Video games / console	2.3
TV	-
Total	37.3

S2.3.4 WEEE end-refining

Umicore recovers gallium from CIGS scrap in special recycling routes only (Oosterhof, 2011). Other end-refiners like Boliden, Aurubis or Remondis generally do not recover gallium, neither from PCB nor from LED. CIGS are not part of this study. Due to this, 100 % of gallium from WEEE is lost in the end-refining.

S2.4 Qualitative material flow models

S2.4.1 Qualitative material flow model of the system “production”

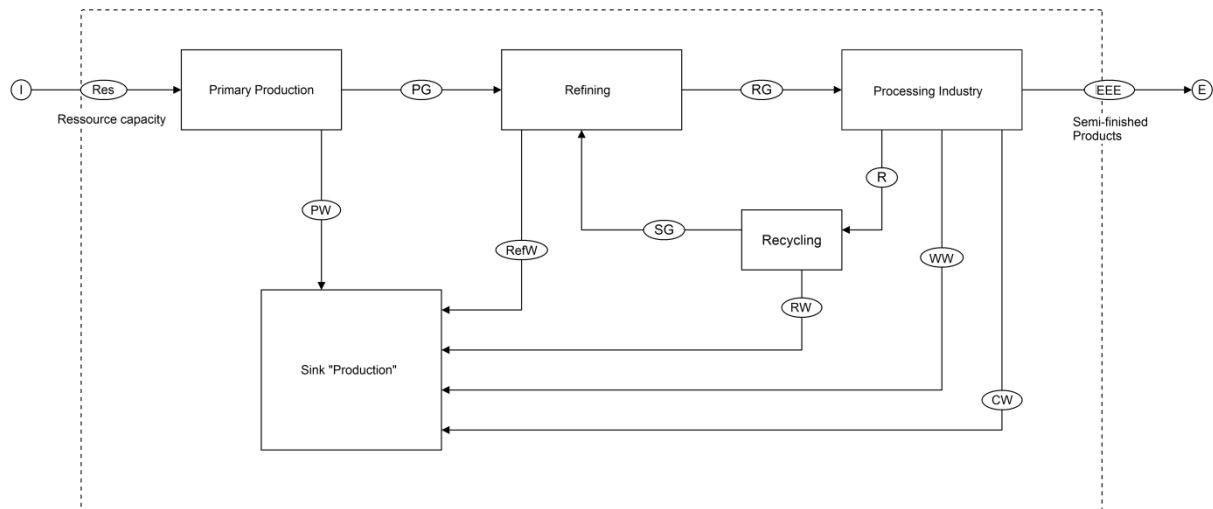


Figure 1: Qualitative material flow model of the gallium “production” system (PG: primary Ga, RG: refined Ga, R: Recyclate, SG: secondary Ga, PW: primary production waste, RefW: refining waste, RW: recycling Waste, WW: production waste (Wafer), CW: production waste (chip))

S2.4.2 Qualitative material flow model of the sub-system “processing industry”

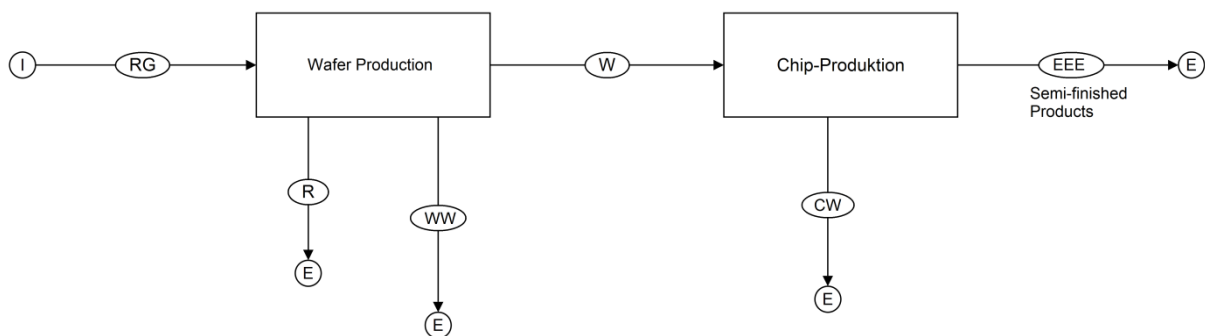


Figure 2: Detailed scheme of the sub-system “processing industry” (RG: refined Ga, W: wafer, R: recyclate, WW: wafer waste, CW: chip waste)

S2.4.3 Qualitative material flow model of the system “waste management”

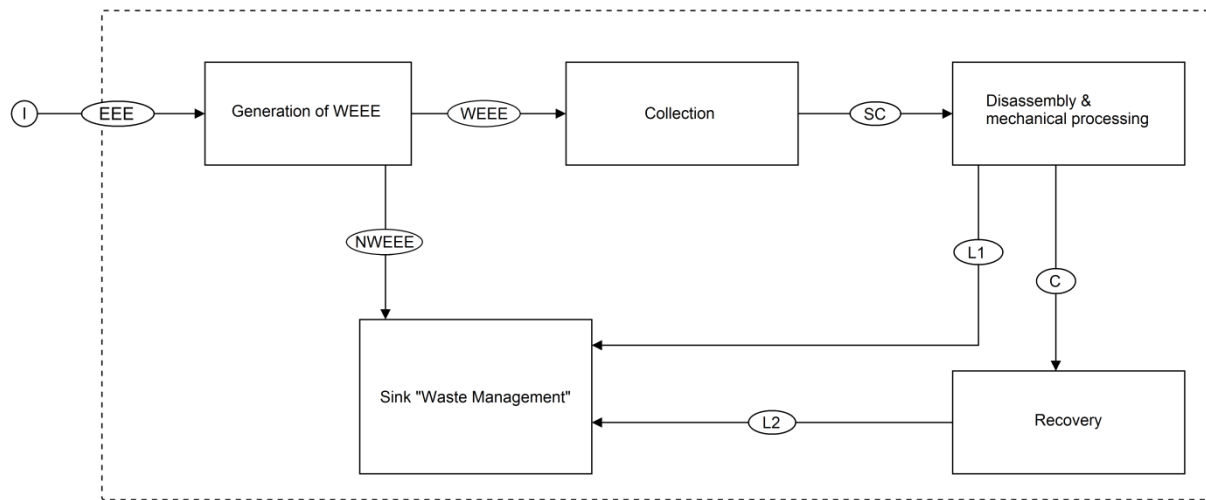


Figure 3: Qualitative material flow model of the system "waste management"(NWEEE: not collected WEEE, SC: separate collection, C: concentrate, L1 and L2: losses)

S3 Thermal treatment of IC

S3.1 Example pictures of IC chips

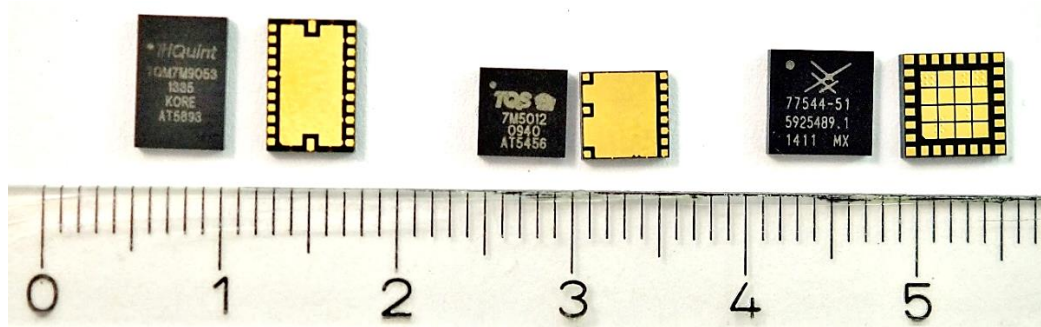


Figure 4: Overview of investigated chip types (from left to right: Type 1, Type 2, Type 3)

S3.2 Weights of IC samples

Table 20: Weights of investigated chips

Chip Nr.°	Weight		
	[mg]		
	Type 1	Type 2	Type 3
1	88.3	61.2	81.4
2	84.4	60.8	81.5
3	85.9	60.6	-
4	84.3	60.8	-
Total	337.9	243.4	162.9

Table 21: Weight losses of chips after thermal treatment

Chip Nr.°	Losses due to thermal treatment		
	[%]		
	Type 1	Type 2	Type 3
1	14.4	8.7	11.8
2	14.8	8.7	11.4
3	14.1	8.6	-
4	14.5	8.9	
Average	14.4	8.7	11.6

S3.3 Output fractions from thermal treatment

The thermally treated IC were separated into 4 different fractions: 1) isolation with copper contacts, 2) metal plate (back of chip), 3) electronic components (chip), 4) miscellaneous

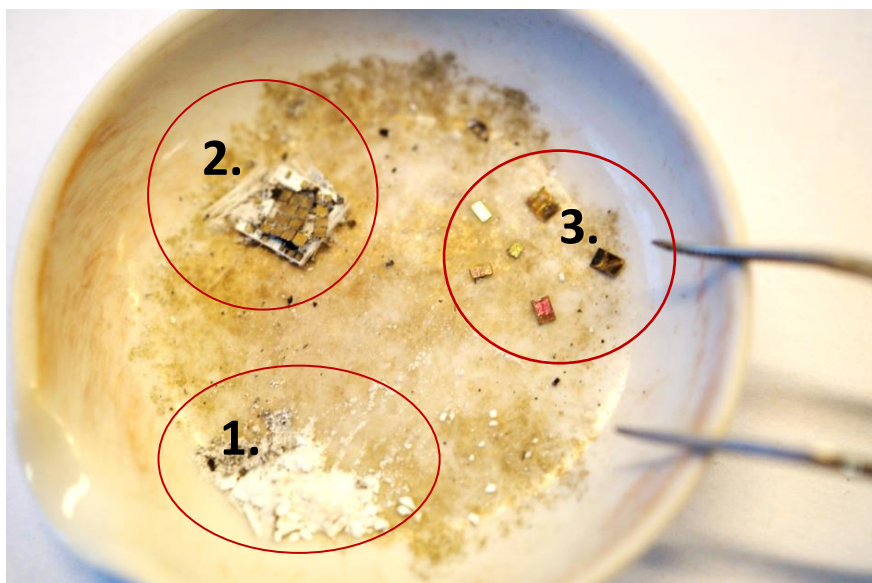


Figure 5: Liberated and sorted components of a high-frequency radio chip after thermal treatment

S3.4 Chemical analyses of output fractions

Table 22: Number of XRF measurements

Probe	Cu	Ga	Ge	As	Ag	Au
T1-2	5	-	-	-	-	5
T1-4	5	-	-	5	5	-
T1-1	5	5	-	5	4	4
T1-3	5	5	5	5	-	5
T2-2	5	-	5	-	5	5
T2-3	5	5	5	5	5	5
T2-1	5	5	-	5	5	5
T2-4	5	-	5	-	5	5
T3-3.1	5	-	5	-	5	5
T3-3.2	5	-	5	5	5	5
T3-3.3	5	5	5	5	5	5
T3-3.4	5	-	2	-	3	5
T3-3.5	5	-	-	5	2	5
T3-3.6	5	5	5	5	5	5
T3-3.7	4	-	-	5	5	5

Table 23: Chemical analyses results of all IC fractions after thermal treatment

Sample	Cu	Ga	Ge	As	Ag	Au
	[mg/g]					
T1-2	684±10	-	-	-	-	14±0.3
T1-4	260±4	-	-	0.7±0.1	3±0.1	-
T1-1	200±30	1±0.5	-	2.2±0.2	0.14±0.03	0.5±0.4
T1-3	19±2	250±11	3±0.3	200±12	-	110±8
T2-2	430±6	-	4±0.6	-	1.2±0.1	110±3
T2-3	115±3	340±9	2.1±0.2	230±8	2±0.1	90±4
T2-1	80±10	1±0.2	-	2.2±0.1	1.5±0.3	1.5±1
T2-4	500±10	-	1.6±0.2	-	4.1±0.1	46±2.5
T3-3.1	0.6±0.07	-	1.5±0.1	-	100±7	55±2.3
T3-3.2	60±30	320±4	3.5±1.4	250±5	17±40	140±55
T3-3.3	500±11	-	6±0.3	2.1±0.3	6.5±1	120±6
T3-3.4	700±20	-	1.6±0.1	-	-	40±4
T3-3.5	140±53	-	-	0.4±0.1	2±0.02	5±0.2
T3-3.6	14±1.3	310±8	3±0.1	280±9	17±1	100±8
T3-3.7	0.42±0.01	-	-	0.7±0.1	70±3	26±0.6

Table 24: Cu, Ga, Ge, As, Ag, and Au in mg

Sample	Cu	Ga	Ge	As	Ag	Au
	[mg]					
T1-2	12.9	-	-	-	-	0.03
T1-4	5	-	-	0.01	0.06	-
T1-1	43	0.2	-	0.5	0.03	0.1
T1-3	0.4	5.2	0.06	4.1	-	2.3
T2-2	16	-	0.2	-	0.05	4
T2-3	1.8	5.4	0.03	3.7	0.03	1.4
T2-1	5.9	0.08	-	0.2	0.1	0.1
T2-4	14.8	-	0.05	-	0.1	1.4
T3-3.1	0.001	-	0.004	-	0.2	0.1
T3-3.2	0.2	1	0.01	0.7	0.05	0.4
T3-3.3	1.4	-	0.02	0.01	0.02	0.4
T3-3.4	8.6	-	0.02	-	-	0.5
T3-3.5	15.6	-	-	0.04	0.2	0.6
T3-3.6	0.04	0.8	0.01	0.8	0.05	0.3
T3-3.7	0.001	-	-	0.001	0.1	0.03
Total type 1 (n=4)	61.2	5.4	0.06	4.6	0.09	2.4
Total type 2 (n=3)	38.6	5.5	0.2	3.8	0.3	7
Total type 3 (n=2)	25.7	1.8	0.06	1.6	0.6	2.3
Average type 1 (n=4)	15.3	1.3	<0.1	1.2	<0.1	0.6
Average type 2 (n=3)	12.9	1.8	<0.1	1.3	0.1	2.3
Average type 3 (n=2)	12.9	0.9	<0.1	0.8	0.3	1.2

S4 Analysis of recycling barriers

Table 25: Basis elements for a barrier analysis to be investigated

Table 29. Basis elements for a barrier analysis to be investigated						
Product related analysis	Product design			Material composition		
	Change in product design	Available product information or labeling	Dissipation through product design	Recycling conflict with other valuable or relevant materials	Complexity of material composition	Recycling conflicts with contaminants
Recycling chain related analysis	Collection of end-of-life devices					
	Collection rates of relevant devices			Trends on collection rates		
	Pre-processing			End-processing		
	Mechanical liberation of target materials	Availability of identification and sorting technology	Complexity of metallurgical liberation	Separability after liberation	End-refining channels available	
Economy related analysis	Economy					
	Financial drivers for the recovery of target material			Energy consumption and waste water/residues production		

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